

Department of Computer Science and Engineering

Academic Year 2023– 2024 (Odd Semester)

Degree, Semester & Branch: VII Semester – B.E. Computer Science and Engineering

Course Code & Title: CS8792 Cryptography and Network Security

Name of the Faculty member (s): Dr.R.Venkatesh, ASP/CSE

Innovative Practice Description

- **Unit / Topic: Unit V / Malicious software: viruses**

- **Course Outcome: CO5**

- **Topic Learning Outcome: TL18**

- **Activity Chosen: Flipped Classroom**

- **Justification:**

Flipping the classroom is a pedagogical approach that reverses the conventional teaching method. In this method, the traditional in-person teaching is restructured to cater to the unique learning requirements of individual students. Students take charge of their learning by independently studying course materials before attending class, such as reading materials and pre-recorded video lectures. This approach allows educators to reimagine in-class activities, which may include interactive problem-solving sessions, aimed at keeping students actively involved in the subject matter.

Time Allotted for the Activity: 20 Minutes

- **Details of the Implementation:**

- **Identify Learning Objectives:** Determine the specific learning objectives for the topic to flip. For example, students should understand the types of viruses that target cryptographic systems and their potential impact.
- **Select Learning Materials:** Identify readings and resources related to the topic. These may include academic articles, research papers, or case studies on malware and cryptography and these materials are shared with the students. Choose or create video lectures or tutorials that explain the concepts of viruses in cryptography.
- **Assign Pre-Class Work:** Provide students with the pre-class assignments, such as reading topics, well in advance of the in-person class session.
- **In-Class Activities:** Plan interactive activities that explore real-world examples of malware attacks on cryptographic systems.
Engage students in discussions and case studies that highlight the importance of security measures to protect against malware in cryptographic applications.

- **Evaluate:** Faculty have prepared few questions related to the content shared and ask those questions in the class session and make the students to write/present in class. The Fig.1 images shows the student Rajasekaran T and Aananda Ganesan R explaining the self-explored concepts on Viruses.
- Make the students to find out the answer by their own by learning

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO7	PO8	PO9	PO10	PSO2
CO2	3	2	2	1	1	1	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO2
Justification for correlation	Apply basic Knowledge and fundamentals in Malicious software: viruses	Identify the need of Malicious software: viruses prevention and detection	Able to find out the mechanism for the viruses prevention & avoidance	As the student involved in the activity as an individual, the individual work skill improved	Communicate / share the ideas with other students in written and oral presentation	The problem solving skill earned through this activity helps the students in solving cyber security problems

- **Images / Screenshot of the practice:**



Figure.1 – Screenshots of the student's activities in the Flipped classroom innovative practice

• **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

- Student felt good, since they can study at their own pace/time.
- They felt that through such learning, they can explore more.

❖ ***Benefit of the practice:***

- More one-to-one time between teacher and student.
- More collaboration time for students.
- Students learn at their own pace.
- Practical things – like missing class due to illness – become less problematic.
- It encourages students to come to class prepared.

❖ ***Challenges faced in implementation:***

- Relies on student preparation – few students did not come prepared.
- The depth of the subject can be dictated by the student themselves or the group the student is working with.
- The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class

References:

- ❖ <https://www.teachthought.com/learning/the-definition-of-the-flippedclassroom/#:~:text=A%20flipped%20classroom%20is%20a,the%20students%20independently%20at%20home.>
- ❖ https://en.wikipedia.org/wiki/Flipped_classroom
- ❖ youtube.com/watch?v=BCIxikOq73Q
- ❖ <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-whatwhy-and-how-to-implement-a-flipped-classroom-model>

Signature of Faculty Member

HOD-CSE